ECO-TRACK: Trash Classification and Face Detection Powered by Computer Vision

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Abstract. Urban waste pollution often results from inadequate collection and sorting practices. We propose ECO–TRACK, a smart waste management system that leverages IoT, embedded systems, and artificial intelligence. Smart bins equipped with ESP32 microcontrollers, ultrasonic sensors, and cameras classify waste in real time using convolutional neural networks (CNNs) and sort it via a servo mechanism. A webcam can capture images of individuals littering, and face recognition is employed to identify repeat offenders under privacy-by-design safeguards. Collected data are transmitted to a central server, where optimized collection routes are computed using a modified Dijkstra algorithm with the OpenStreetMap (OSM) API, prioritizing full bins to reduce trips, fuel, and emissions. The system is designed to handle diverse waste types under varied conditions and to accurately separate recyclable, compostable, and non-recyclable categories. Overall, ECO–TRACK improves sorting, monitoring, and collection efficiency in a scalable, intelligent, and environmentally sustainable manner.

Keywords: Smart Waste Management · Trash Classification · CNN · Face Recognition · Route Optimization

1 Introduction

ECO–TRACK merges sustainability with smart technologies to modernize urban waste management. In many cities, population growth and consumption patterns have increased waste production. Traditional approaches (manual sorting, static routes) are increasingly ineffective. Improper classification leads to low recycling rates, landfill growth, and environmental degradation.

Real world problem. Urban waste systems face pressure from rapid urbanization and changing consumption. Manual sorting is labor intensive and error prone; static schedules waste fuel and cause overflow or missed pickups.

Importance. Inefficiencies contribute to pollution, greenhouse gases, and public health risks. Many smart systems rely heavily on cloud resources that can be costly or inaccessible; there is a need for decentralized, adaptive solutions.

Gap. Prior AI/IoT solutions often depend on cloud inference, have limited classification scope, or omit end-to-end automation (physical sorting and dynamic

routing). Few are low-cost, real-time, and autonomous for infrastructure-limited settings.
Problem statement. We present an edge-computing, AI-driven solution that performs real-time waste classification and sorting, monitors bin status, and optimizes routes dynamically to reduce inefficiency and promote recycling.

ECO–TRACK centers on the ESP32-CAM for on device image capture and lightweight CNN based classification of recyclable/compost/trash with low latency and energy use. Servo motors actuate physical sorting, and ultrasonic sensors track fill levels. A central server computes dynamic routes via Dijkstra and OSM. A facial recognition module (event-triggered) can identify repeat offenders to encourage accountability, implemented with data minimization and safeguards.
Contributions. Our contributions are: (i) a fully integrated, low-cost edge system combining on-device CNN classification (ESP32-CAM) with mechanical sorting; (ii) online bin monitoring with ultrasonic sensing and dynamic OSM-based route optimization under capacity constraints; (iii) an accountability module with event-triggered face recognition designed with privacy-by-design; and (iv) an empirical evaluation on two multi-class datasets with ablations and error analysis.
Paper organization. Section 2 reviews related work. Section 3 details materials and methods. Section 4 presents experiments and results. Section 5 concludes and outlines future work.

2 Related Work

Automated trash classification has progressed steadily from early convolutional neural networks (CNNs) on the TrashNet dataset, such as the work of Alwateer et al. (87% accuracy), to more advanced transfer learning approaches on lightweight architectures like MobileNet, which enabled efficient deployment on edge devices (Rad et al., 92.3%). Other studies have emphasized both accuracy and explainability, for instance ResNet50 combined with Grad-CAM visualization techniques (Bai et al., 95.4%), while lightweight variants such as EfficientNet-B0 have been adapted specifically for smart bin contexts, achieving ≈93.8% performance with lower computational demand.

In parallel, advances in face recognition have expanded the accountability dimension of smart waste management. Widely adopted deep models such as FaceNet, ArcFace, YOLOFace, and InsightFace have demonstrated robust recognition performance across unconstrained environments, making them suitable for monitoring and deterrence in public spaces. Other complementary contributions, such as VisNet for transformation-invariant object recognition and CNN-based mask detection systems (97.05% accuracy in real time), highlight the versatility of vision models across domains that share challenges with waste monitoring.

Finally, in the domain of collection logistics, operations research approaches such as Lagrangian-relaxation-based routing have shown measurable improvements in reducing collection costs and travel distances. Together, these strands of research have provided strong building blocks. However, prior works often optimize isolated components—classification *or* recognition *or* routing—without offering a unified solution. ECO–TRACK addresses this gap by combining embedded CNN-based classification, event-triggered face recognition, and dynamic route optimization into a single, low-cost, and scalable framework [5,15,14,3,2,9,8,6,1,4,10]. A concise comparison of representative studies is provided in Table 1.

Table 1: Comparison of related studies (abridged).

Study	**Approach / Model**	**Performance**	**Key Contribution**
Alwateer et al. [13]	CNN on TrashNet	87% acc.	Early CNN; class imbalance mitigation
Rad et al. [13]	MobileNet TL	92.3% acc.	Edge-optimized transfer learning
Thung & Yang [11]	CNN + image proc.	88.7% F1	Hybrid features for classification
Bai et al. [3]	ResNet50 + Aug + Grad-CAM	95.4% acc.	Accuracy + interpretability
Deep FR models [4]	FaceNet/YOLOFace	High (varies)	Littering accountability

3 Materials and Methods

We outline dataset preparation, model architecture/training, system integration, route optimization, and model selection.

3.1 Dataset Preparation

Two Roboflow datasets were used: the first [7] with *recyclable* and *trash*; the second [12] adds *compost*. Images span varied environments, lighting, occlusions, and orientations. Each image includes YOLO/COCO format bounding boxes. This diversity supports robust generalization. A compact summary of Dataset 1 is given in Table 2.

Table 2: Dataset 1 (Roboflow) summary.

Property	Details
#Images	3,200
Classes	Recyclable; Trash
Split (Train/Val/Test)	70 / 15 / 15
Distribution (%)	Recyclable 45%; Trash 55%

Figure 1 presents sample images from the dataset used in this study, illustrating the three primary waste classification categories: recyclable, trash, and compost.

These examples highlight the diversity and complexity of visual features within each class, including variations in lighting, occlusion, background clutter, and object deformations.

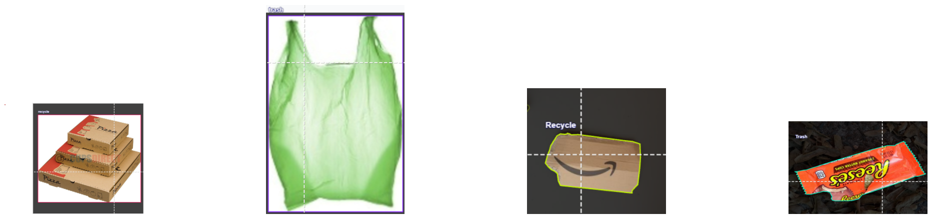


Fig. 1: Representative images across classes: recyclable, trash, and compost.

3.2 Model Architecture and Training

We used YOLOv8 for high-performance detection (anchor-free with enhanced backbone and auto-label assignment). Preprocessing includes resizing to 640×640, normalization, and augmentations such as flip, scale, brightness, and rotation. Transfer learning is applied from COCO weights, followed by fine-tuning on our datasets with class-specific bounding boxes. Mean Average Precision (mAP) is reported at IoU thresholds (mAP@0.5 and mAP@0.5:0.95), where IoU is Intersection over Union. The main components are illustrated in Figure 2 for trash classification and in Figure 3 for face detection.

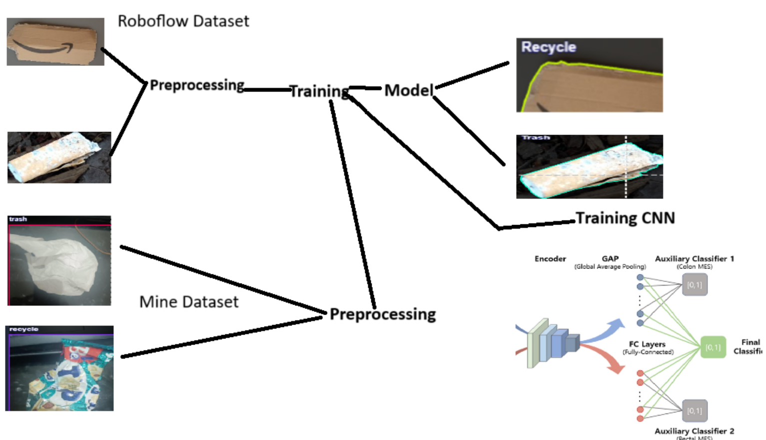


Fig. 2: Trash classification model architecture overview.

The waste classification module (Figure 2) is complemented by a face detection component, shown next, which enables accountability and user monitoring within the system.

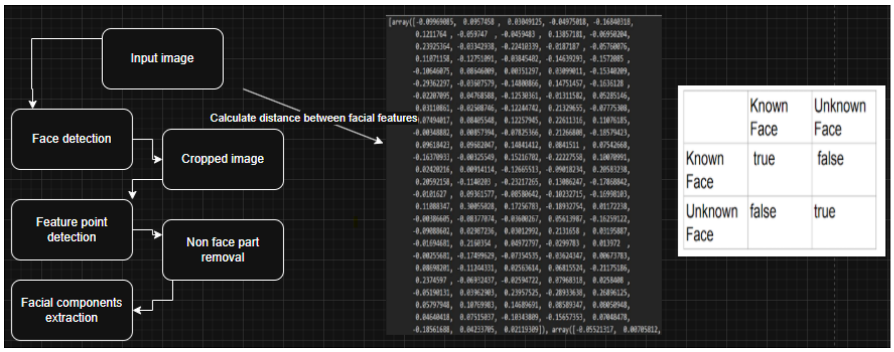


Fig. 3: Face detection model.

3.3 System Integration: Hardware–Software Pipeline

The end-to-end hardware/software flow of ECO–TRACK is shown in Figure 4. The ESP32-CAM captures an item, runs a compact CNN locally (quantized for TensorFlow Lite for Microcontrollers, TFLM), and triggers the servo to sort. Ultrasonic sensors publish fill-level events; when thresholds are exceeded, bin states are sent to the server. The server aggregates states, queries OSM, and computes dynamic routes (Section 3.4). Accountability images, when generated, are encrypted, access-controlled, and retained briefly (see Ethics).

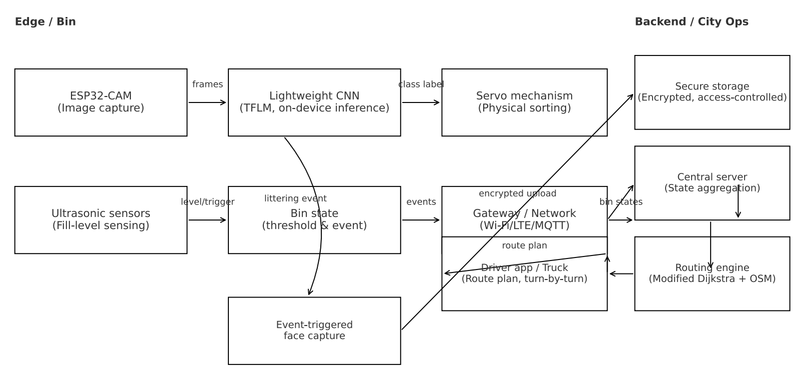


Fig. 4: Hardware–software pipeline of ECO–TRACK.

3.4 Route Optimization

Let $G = (V, E)$ be the OSM road graph with nonnegative travel costs c_{uv}. We compute a depot-rooted route minimizing $\sum c_{uv}$ subject to: (i) visiting bins with fill-level $\geq \tau$, (ii) truck capacity constraint $\sum_i w_i \leq C$ (volume estimate w_i, capacity C), and (iii) return to depot. We use a modified Dijkstra to obtain a feasible near optimal sequence by capacity-aware pruning and edge reweighting with bin priority (fill-level and organic content). This heuristic yields large fuel/time gains and supports online updates [10].

3.5 Model Selection: YOLOv8 vs ResNet

To motivate the deployment choice, Table 3 contrasts YOLOv8 and ResNet across dimensions such as primary task, output type, computational overhead, and feasibility on edge devices. While YOLOv8 excels in object detection with high accuracy, its resource demands make it less suitable for low-power microcontrollers. In contrast, ResNet offers a classification-focused backbone with modular depth and established transfer learning benefits, which can be adapted more efficiently for constrained hardware. We therefore derive a lightweight CNN inspired by ResNet principles, incorporating model quantization and pruning to reduce memory footprint and energy cost, and specifically tailoring deployment for ESP32-CAM, ensuring real-time inference under stringent edge constraints.

Table 3: Simplified comparison between YOLOv8 and ResNet.

Feature	YOLOv8	ResNet
Primary Task	Object Detection	Image Classification
Output	Boxes + labels	Single class label
Speed	Real-time (with GPU)	Task-dependent
Architecture	Single-stage detector	Deep CNN (residual)
Performance	High mAP	High accuracy
Hardware Needs	High (GPU/TPU)	Moderate–High
Edge Feasibility	Low on MCU	Moderate (with compact CNN)

4 Experiments and Results

We fine-tuned two YOLOv8 models on distinct datasets (two-class and three-class). We report precision, recall, mAP, F1, PR curves, and confusion matrices, and summarize validation performance.

4.1 Training Settings and Metrics

Precision $P = \frac{TP}{TP+FP}$, Recall $R = \frac{TP}{TP+FN}$, and F1 score $= 2 \cdot \frac{PR}{P+R}$ are the main evaluation metrics used. Additionally, we report mean Average Precision (mAP) at IoU thresholds of 0.5 and the range 0.5:0.95. The training hyperparameters are listed in Table 4.

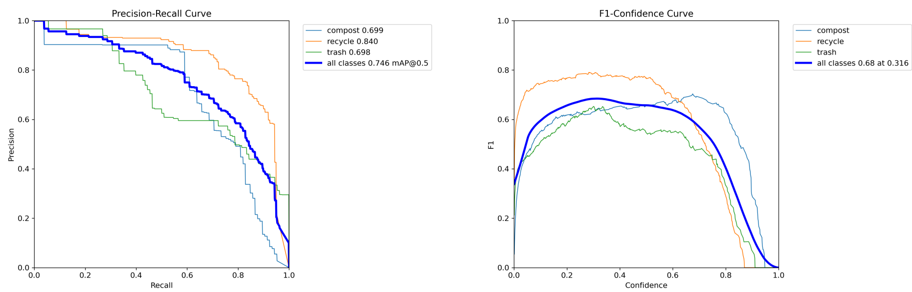
Table 4: Training hyperparameters (classification/detection).

Parameter	Value
Learning Rate	0.001
Optimizer	Adam
Batch Size	32
Epochs	50
Loss	Categorical Cross-Entropy
Activation	ReLU (hidden), Softmax (output)
Early Stopping	Patience 5 (val. loss)

4.2 Performance

Figure 5 provides a detailed view of the validation performance through Precision–Recall (PR) curves for each class as well as the F1–confidence curve, which is particularly useful for identifying an operating threshold that balances precision and recall in practical deployment. These curves highlight how well the model is able to maintain high recall without sacrificing too much precision and vice versa, offering a deeper understanding of trade-offs across different thresholds. In addition, Figure 6 illustrates the confusion matrix for the three-class problem, revealing the dominant sources of misclassification, such as the overlap between compost and trash due to their similar visual features. Finally, Table 5 summarizes the quantitative results by reporting precision, recall, and mean Average Precision (mAP) scores at different IoU thresholds, thereby providing a comprehensive numerical benchmark of the model's effectiveness across all categories.

Together, these results establish both the strengths of the system in handling recyclable items and the challenges that remain in reliably distinguishing between compost and trash in real-world conditions. They also demonstrate that while the model performs competitively relative to prior work, further refinements are required to close the gap with state-of-the-art accuracy levels. These insights are crucial for guiding future improvements, such as integrating multi-modal sensing (e.g., weight, gas, or RFID-based inputs) to complement vision-based recognition and reduce ambiguity in visually similar waste types. Moreover, they emphasize the value of advanced augmentation strategies and domain adaptation techniques to enhance robustness under varied lighting conditions, occlusions, and environmental noise. By analyzing the interplay between qualitative error patterns and quantitative performance metrics, these evaluations not only validate the feasibility of ECO–TRACK in constrained edge settings but also highlight concrete directions for achieving more reliable, scalable, and environmentally impactful deployments.



(a) PR curves (validation). (b) F1–confidence curve.

Fig. 5: Validation PR curves and F1–confidence curve.

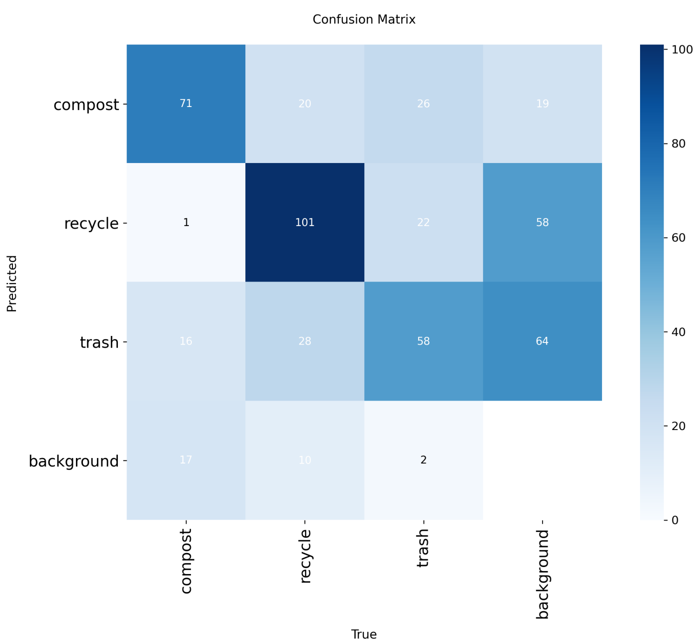


Fig. 6: Confusion matrix for the three-class model.

Table 5 reports validation results, with the *recycle* class showing the best precision (0.762) and recall (0.806). Compost and trash achieve lower scores due to visual similarity, though recall remains solid across all classes, ensuring most items are detected despite occasional mislabels.

Table 5: Validation metrics (Dataset 2, three classes).

Class	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Compost	0.593	0.705	0.699	0.652
Recycle	0.762	0.806	0.840	0.723
Trash	0.576	0.722	0.698	0.660
Overall	0.644	0.744	0.746	0.678

Interpretation and Discussion. The proposed YOLOv8 model achieves an overall mAP@0.5 of 74.6%, which demonstrates strong performance given the constraints of edge deployment on low-cost hardware. Among the three categories, the recycle class performs the best, reaching a precision of 76.2% and a recall of 80.6%, indicating that recyclable materials are both consistently detected and rarely missed. In contrast, the compost and trash classes exhibit comparatively lower precision, largely due to their overlapping visual characteristics—such as similar textures, colors, and degrees of decomposition—that make them harder to distinguish reliably. Despite this, recall values remain higher than precision across all categories, meaning the model favors completeness of detection (minimizing false negatives) at the expense of occasional false positives, which is generally preferable for waste-sorting scenarios where missed detections can undermine sustainability goals.

Ethical and Privacy Considerations

The implementation of facial recognition raises important ethical questions. ECO–TRACK follows privacy-by-design and data minimization: (i) event-triggered capture (e.g., littering) rather than continuous recording; (ii) short retention windows with deletion of non-violating footage; (iii) encrypted storage and transport with access restriction and audit logs; and (iv) visible signage/notice where applicable. Only images relevant to repeated violations are retained, and identities are not shared beyond authorized municipal stakeholders. Deployments should align with applicable data protection regulations (e.g., GDPR-like principles), including lawful-basis assessment and Data Protection Impact Assessment (DPIA). Misidentification and bias risks are mitigated through calibrated thresholds, periodic human-in-the-loop review, and fairness audits on diverse datasets.

Limitations

The evaluation is image-centric; performance can degrade under extreme lighting or heavy occlusion. Compost vs. trash confusion persists. ESP32-CAM constraints limit model depth and input resolution. Future work will examine sensor fusion, adaptive thresholds, and cross-city generalization.

Reproducibility

Training configurations and model checkpoints can be provided to enable exact reruns of reported experiments.

5 Conclusion and Future Work

ECO–TRACK combines embedded sensors, computer vision, and AI to automate sorting and optimize collection. A compact CNN on ESP32-CAM performs real-time classification (recyclable/compost/trash), ultrasonic sensors track fill levels, and a modified Dijkstra algorithm computes efficient routes over OSM, reducing fuel use and emissions. An event-triggered face-recognition module supports accountability under explicit privacy safeguards. Results show reliable performance for real-world deployment.

Future work includes fully autonomous smart bins and robotic collection vehicles, larger and more diverse datasets, city-scale analytics, and improved edge performance. We also plan to explore federated learning for privacy-preserving updates across distributed bins.

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